

Final Project: Predicting Hospital Readmissions

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Hospital readmissions within 30 days remain one of the most persistent cost drivers and quality challenges in the U.S. healthcare system. Under the Centers for Medicare & Medicaid Services (CMS) Hospital Readmissions Reduction Program (HRRP), hospitals face financial penalties when readmission rates exceed national benchmarks. To address this challenge, our team developed a machine-learning–based risk prediction framework designed to identify patients at elevated risk of being readmitted within 30 days of discharge.

The project utilized the UCI Diabetes 130-US Hospitals dataset, consisting of 69,990 cleaned patient encounters. As noted in the dataset documentation, *“the provided data suffers from significant class imbalance, as ~9% of records belong to the positive class.”* This imbalance presents a substantial modeling challenge, as rare clinical events are inherently difficult to predict. Data preparation included extensive cleaning, missing-value treatment, deduplication, and encoding of demographic, clinical, and medication-related variables. Numerical features were standardized. To address imbalance, SMOTE oversampling was applied to the training set.

As specified by the CMS HRRP policy, the penalty that hospitals with excessive readmissions can be subjected to is a reduction in their Medicare payments ranging between 0% and 3%¹ per case. Since HealthBridge Medical Center has a readmission rate of 18.2%, which is higher than the national average of 15.5%², it falls within the penalty range. According to in-house estimates, HealthBridge may suffer a penalty of up to \$2.4 million in annual payments under the HRRP scheme and incur an additional cost of \$6.1 million in FY2024.

Three supervised learning algorithms were evaluated: logistic regression, random forest, and gradient boosting (XGBoost). Despite robust preprocessing, all models struggled to achieve strong discriminatory performance. The random forest model identified clinically intuitive predictors—such as length of stay, discharge disposition, and prior utilization—but its predictive

power remained limited. XGBoost achieved the highest ROC-AUC at 0.591, while logistic regression and random forest performed slightly lower. Precision-recall metrics further highlighted the difficulty of identifying true high-risk patients, with all models demonstrating low recall for the positive class.

Although the models do not yet support autonomous clinical decision-making, they provide valuable directional insight. The analysis reinforces that discharge planning quality, length of stay, and post-discharge disposition meaningfully influence readmission risk. These findings can guide care-management teams in prioritizing follow-up interventions, even if the predictive model itself is not yet deployment-ready.

For future development, several enhancements are recommended. First, incorporating richer clinical variables—such as comorbidity indices, laboratory trajectories, and historical utilization patterns—may improve signal strength. Second, alternative imbalance-handling strategies (cost-sensitive learning, or ensemble methods tailored for rare events) will be explored. Finally, integrating real-time EHR data and social determinants of health may significantly improve predictive accuracy.

In conclusion, this project establishes a foundational machine-learning framework for readmission risk prediction. While current model performance is modest, the analytical process, feature engineering pipeline, and evaluation methodology provide a strong platform for future refinement. With additional data and enhanced modeling techniques, this approach has the potential to support hospitals nationwide in reducing avoidable readmissions and improving patient outcomes.

References

¹Centers for Medicare & Medicaid Services. (2024). Hospital Readmissions Reduction Program (HRRP). <https://www.cms.gov/medicare/quality/value-based-programs/hospital-readmissions>

UCI Diabetes 130-US Hospitals (1999–2008)

<https://archive.ics.uci.edu/dataset/296/diabetes+130-us+hospitals+for+years+1999-2008>

²Unplanned Hospital Visits – National <https://data.cms.gov/provider-data/dataset/cvcs-xecj>

Attribution

Project review feedback assistance provided by Claude (Anthropic, 2025). <https://claude.ai>"

Microsoft Copilot. (2026, April 19). *Assistance with review of Readmission Risk Modeling and executive summary*. Microsoft Copilot (AI companion). Retrieved from

<https://copilot.microsoft.com>